International Journal of Disaster Risk Science (2025) 16:376–391
https://doi.org/10.1007/s13753-025-00644-3

www.ijdrs.com
www.springer.com/13753

ARTICLE

Digital Platform Empowering China-Pakistan Economic Corridor Disaster Management: Enhancing Data Integration, Resilience, and Policy Innovation

Juanle Wang[1,3,4] · Kun Bu[2] · Congrong Li[1,4] · Yuelei Yuan[1] · Xinrong Yan[1] · Chen Xu[1] · Meer Muhammad Sajjad[1,3] · Lina Hao[1,3] · Jiazhuo Liu[1] · Xiaodong Min[1]

Accepted: 26 May 2025 / Published online: 24 June 2025
© The Author(s) 2025

Abstract
The China-Pakistan Economic Corridor (CPEC) region faces persistent threats from natural hazards and disasters, posing significant challenges to sustainable development. While digital technologies and big data offer new solutions for disaster management, imbalanced digital resources and inadequate data management systems hinder their full potential. This study proposed the Digital CPEC platform, an open technology solution (using, for example, Python, PostgreSQL, MapServer) developed to share disaster-related open data in a fully digital format, thereby enhancing data accessibility and lowering economic and technical barriers. The platform offers a scalable and replicable model for strengthening disaster resilience in underdeveloped and transboundary regions worldwide, driving innovation in policy assurance, data integration, software support, and customized deployment. To ensure data integrity and security, four disaster management standards and a structured data integration framework were established, consolidating resources from seven thematic data corridors spanning natural hazard, ecological environment, and socioeconomic fields. Additionally, 18 metadata indicators and 38 identifiers were introduced into the metadata schema to enhance data quality and usability, alongside specialized analytical tools (for example, Echarts, area calculation) that improve user interaction. Its practical applications in pre-disaster risk monitoring, early warning coordination, and post-disaster recovery assessment demonstrate its strong knowledge-driven policy insights for environmental sustainability throughout the entire disaster management cycle at both the regional and national levels.

Keywords China-Pakistan Economic Corridor · Data sharing · Digital platform · Disaster management · Regional sustainable development

1 Introduction

The China-Pakistan Economic Corridor (CPEC), one of the six major economic corridors under the Belt and Road Initiative (BRI), stretches over 3000 km from Kashgar, Xinjiang, China, to Gwadar Port in Pakistan (Bhattacharjee 2015). This corridor integrates the Silk Road Economic Belt in the north with the 21st Century Maritime Silk Road in the south, serving as a flagship project for the BRI, playing a pivotal role in fostering cooperation between China and Pakistan in energy, transportation, and economic development. Since the signing of a long-term cooperation agreement in 2015, the corridor has seen investments valued at USD 65 billion by 2022 (Shahzad 2022).

Environmental governance, climate-resilient development, and environmental sustainability are crucial for the long-term success of large-scale infrastructure projects. In this context,

✉ Congrong Li
licongrong@igsnrr.ac.cn

1 State Key Laboratory of Resources and Environmental Information System, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing 100101, China

2 Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun 130102, China

3 University of Chinese Academy of Sciences, Beijing 100049, China

4 Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application, Nanjing 210023, China

Waheed et al. (2024) assessed Pakistan's environmental governance, emphasizing the importance of transparency and access to reliable information for effective management. Latief et al. (2024) analyzed the relationship between sustainable tourism, renewable energy, technological innovation, and environmental quality in BRI countries, highlighting that promoting environmentally friendly growth strategies (such as sustainable tourism) and fostering technological innovation are key to global cooperation in addressing climate change and advancing environmental sustainability. Butt et al. (2024) explored CPEC's role in Pakistan's coastal development and blue economy, proposing strategies—CPEC "1+5" model and Maritime Special Economic Zones—to maximize benefits while minimizing environmental risks, encompass natural and cultural heritage preservation, community engagement, responsible tourism, eco-friendly infrastructure, destination branding, and local collaboration. Waheed, Kousar et al. (2025) evaluated CPEC's sustainability using a SWOT (strengths, weaknesses, opportunities, and threats) framework, highlighting technological inefficiencies and overexploitation of natural resources as key internal weaknesses and external threats. Collectively, these studies highlight the need for greater data transparency and accessibility, the promotion of integrating sustainability principles, and fostering technological innovation to ensure CPEC's long-term environmental sustainability and resilience.

The occurrence of natural hazards and disasters has further exacerbated the complexity of environmental governance, climate-resilient development, and environmental sustainability. Pakistan's geographic location, heavy reliance on agriculture and water resources, low adaptive capacity, and inadequate emergency preparedness make it highly vulnerable to climate change (Yamada et al. 2016; Song et al. 2023) and ecological degradation (Phong et al. 2009), posing significant threats to the region (Mueller et al. 2014; Mekonnen and Hoekstra 2016; Su et al. 2016; Mondal et al. 2021; Song et al. 2023). Situated along the geologically and climatically complex Himalayas, Karakoram, and Hindu Kush mountain ranges, the corridor is highly susceptible to diverse natural hazards. According to Pakistan National Disaster Risk Reduction (DRR) Policy 2013, the most frequently occurring disasters in Pakistan include floods, droughts, storms, avalanches, glacial lack outbursts, landslides, earthquakes, and tsunamis (GOP 2013). Over the past two decades, climate-related disasters have caused economic damages exceeding USD 20 billion (Atta Ur and Shaw 2015), while environmental degradation accounted for 9% of Pakistan's GDP in 2015 (up from 6% in 2006) (GOP 2017). Earthquakes and flooding are the most destructive disasters in Pakistan. Between 2001 and 2013, Pakistan reported 25 extreme events, including six earthquakes (74,484 fatalities) and six landslides, along with annual occurrences of floods (5722 fatalities), collectively leading to 80,415 fatalities (Atta Ur and Shaw 2015). The 2022 floods alone directly affected two-thirds of its territory, displaced over 33 million people, destroyed 1.7 million homes, resulting in economic losses at around USD 10 billion, and causing nearly 1,500 fatalities (Otto et al. 2023).

Despite the frequency of these events, Pakistan's climate vulnerability is compounded by weak disaster management systems and policies (Waheed et al. 2021; Waheed et al. 2023; Waheed et al. 2024). Although the National DRR Policy and National Environmental Policy (NEP) outline objectives and strategies addressing climate change, disasters, development, and the environment, they fail to integrate disaster risk reduction (DRR) as a policy objective and lack clearly defined implementation steps regarding "who, what, when, and how." These policies also lack comprehensive monitoring and evaluation mechanisms, standardized methodologies, and well-defined criteria for policy revision, assessment, and outcome measurement, which significantly limit their effectiveness. Weak data collection mechanisms before and after policy implementation further exacerbate these shortcomings (Waheed et al. 2023). These systemic deficiencies have hindered Pakistan's progress in achieving global frameworks such as the Millennium Development Goals (MDGs), the Hyogo Framework for Action 2005–2015 (HFA), the Sendai Framework for Disaster Risk Reduction 2015–2030, and the United Nations Sustainable Development Goals (SDGs) (Waheed et al. 2023).

Digital technologies and big data present transformative potential in addressing these disaster management challenges (Khan et al. 2020), enhancing data integration, resilience, and policy innovation, from pre-disaster mitigation—such as flood simulations guiding infrastructure planning (Fischer-Preßler et al. 2024)—to real-time response and post-disaster recovery (Fujita and Hatayama 2023). Digital disaster management platforms hold strong potential for effective data management. Currently, those developed in Pakistan present significant opportunities and challenges across four key areas.

(1) Internal Strengths: Pakistan launched the Digital Pakistan initiative on 5 December 2019, aiming to foster a tech-driven ecosystem for social welfare and environmental sustainability (Nizam et al. 2020). Building on this effort, the Digital Nation Pakistan Bill, introduced on 16 December 2024, outlines plan to create a secure, inclusive, and interoperable digital public infrastructure. This initiative targets a fully digital society, economy, and governance, promoting innovation, connectivity, and sectoral integration.
(2) Internal Weaknesses: Pakistan faces significant obstacles to digitalization as most of its citizens are unfamiliar with, unable to use, or distrust digital processes, struggles with limited access to modern technologies,

low computer literacy among the workforce in both public and private sectors, and outdated infrastructure (Ahmed 2019; Nadeem et al. 2024). These obstacles hinder the adoption of digital processes, particularly in disaster management and environmental governance.

(3) External Opportunities: Global disaster data platforms like the Emergency Events Database (EM-DAT), Copernicus Emergency Management Service (EMS), Sigma Database, Munich Reinsurance Company, and DesInventar provide valuable resources, but they often fall short in addressing region-specific challenges. Recognizing this, the CPEC data sharing platform for resources, environment, ecology, and disasters was established by the China-Pakistan Joint Research Center on Earth Sciences in 2020. This platform aims to aggregate and integrate multisource data to create long-term spatiotemporal datasets, laying a robust foundation for sustainability assessments, environmental protection, and DRR efforts.

(4) External Threats: The data platforms face significant challenges in data quality control and sharing security (particularly in cross-border data sharing), lacking standardized metadata standards and data management protocols.

In response, this study proposed the Digital CPEC platform—a comprehensive, integrated digital solution aligned with national initiatives and the SDGs to address key challenges in DRR and sustainability, showcases geographic information as well as significant advances in service effectiveness in a fully digitalized view. This platform seeks to bridge the digital divide (*Q1: How can digital technologies and big data be leveraged to build dedicated digital platforms to bridge the digital divide, thus addressing the lack of tailored solutions for disaster risk management (DRM) in the CPEC region?*), enhance data integration (*Q2: How can multisource and multidisciplinary data be integrated to address complex geopolitical and geophysical challenges in regions like CPEC, and contribute to improving disaster resilience through region-specific solutions?*), and enable effective disaster management (*Q3: What standardized frameworks and policies are needed to enable seamless integration of data and foster trust in digital systems, ensuring effective DRM in the CPEC region?*).

Its key objectives are providing a replicable model—data aggregation, open sharing, rapid response, and research promotion—for underdeveloped regions as follows.

(1) Data Aggregation: Establishing standardized technical frameworks for data integration to unify multisource geospatial and multidisciplinary datasets, enabling comprehensive risk assessments and informed decision making tailored to regional needs.

(2) Open Sharing: Developing robust, standardized data-sharing protocols for secure, transparent, and accessible data sharing—with incentives to facilitate cross-border collaboration among stakeholders.

(3) Rapid Emergency Response: Designing specialized modules for real-time disaster data aggregation and analysis, enhancing forecasting capabilities and ensuring timely, coordinated interventions.

(4) Scientific Research Promotion: Supporting interdisciplinary research and dynamic data updates, fostering a comprehensive life-cycle data management system, facilitating a foundation for knowledge-based policy formulation and decision making.

The remainder of this article is organized as follows: Section 2 outlines the platform's design and data management system. Section 3 elaborates on its functionalities, including data aggregation and sharing protocols. Section 4 highlights enhancements in emergency response and research support. Section 5 discusses the platform's contributions and future prospects, followed by conclusions in Sect. 6.

2 Digital China-Pakistan Economic Corridor (CPEC) Platform Design and Data Management

This section is structured as two parts: The first part outlines the design of the Digital CPEC platform, emphasizing that open science and technologies help address digital resource disparities within the CPEC region. The second part focuses on the data management framework, highlighting methods and standards to strengthen data management and bridge gaps in existing systems.

2.1 Digital China-Pakistan Economic Corridor (CPEC) Platform Design and Development

Digital CPEC is designed to support DRR and related scientific research activities within the CPEC region by utilizing rich open technologies, digital datasets, and visual maps. It prioritizes simplification to enhance development efficiency and minimize maintenance costs by adopting modern software development methodologies that promote architectural adaptability and scalability through structured organization and component decoupling. In addition, it adopts the FAIR (Findable, Accessible, Interoperable, and Reusable) (Wilkinson et al. 2016) principles to ensure data usability and transparency.

2.1.1 Platform Design

The Digital CPEC platform is a comprehensive geographic information service system, divided into five primary system modules and seven core functional modules. This integrated modular design ensures the platform's scalability, adaptability, and maintainability, aligning with future technological advancements and evolving user requirements.

Five primary system modules we designed (Fig. 1) include: (1) Portal Website Module (PWM): Serves as the main user interface, facilitating interactions and managing client requests; (2) Data Service Platform Module (DSPM): Handles metadata management, data publication, and user request execution while providing real-time feedback through the main user interface; (3) Scientific Database System Module (SDSM): Stores and manages diverse datasets, including geographic, environmental, socioeconomic, and ecological data; (4) Geographic Spatial Information Module (GSIM): Enables spatial data visualization and analysis, offering map services and geographic information tools through its service interfaces; (5) Front-End Visualization Application Module (FEVAM): Provides a user-friendly interface for resource display, map overlays, and user engagement.

Seven core functional modules we designed include: (1) Data Resource Addition and Update Module, which allows users to easily upload and update data, with metadata editing capabilities for online browsing and seamless integration; (2) Document Addition and Update Module, which supports the upload, parsing, and storage of resource documents, with the ability to add metadata for detailed descriptions and spatial location visualization; (3) Resource Retrieval Module, which uses full-text search technology for fast and accurate data discovery; (4) User Management Module, which manages user permissions, enabling efficient addition, deletion, modification, and querying of user information; (5) Metadata Management and Catalog Service Publication Module, which promotes data sharing and interoperability; (6) Online Map Application Module, which offers functionalities such as real-time geographic location visualization, online map overlays, referencing, and editing with storage capabilities; and (7) Front-End Visualization Application Module, which focuses on user-centric design, and provides directory navigation and filtering tools for quick and efficient data access. This modular design ensures scalability and adaptability, accommodating future technological advancements and evolving user needs, and enabling tailored services.

2.1.2 Platform Development

The platform uses WebGIS (Web-based geographic information system) visualization technology and a cloud GIS architecture to enable online access to diverse data resources, including vector data, raster data, statistical data, and remote sensing imagery. Its development is grounded in open-source technologies such as Python (programming language), PostgreSQL (database management system), and MapServer (open-source development environment). Operating through a centralized Web Service hub, the platform provides interfaces for various services, including Web Map Services (WMS), Web Data Services (WDS), Web Metadata Services (WMTS), and

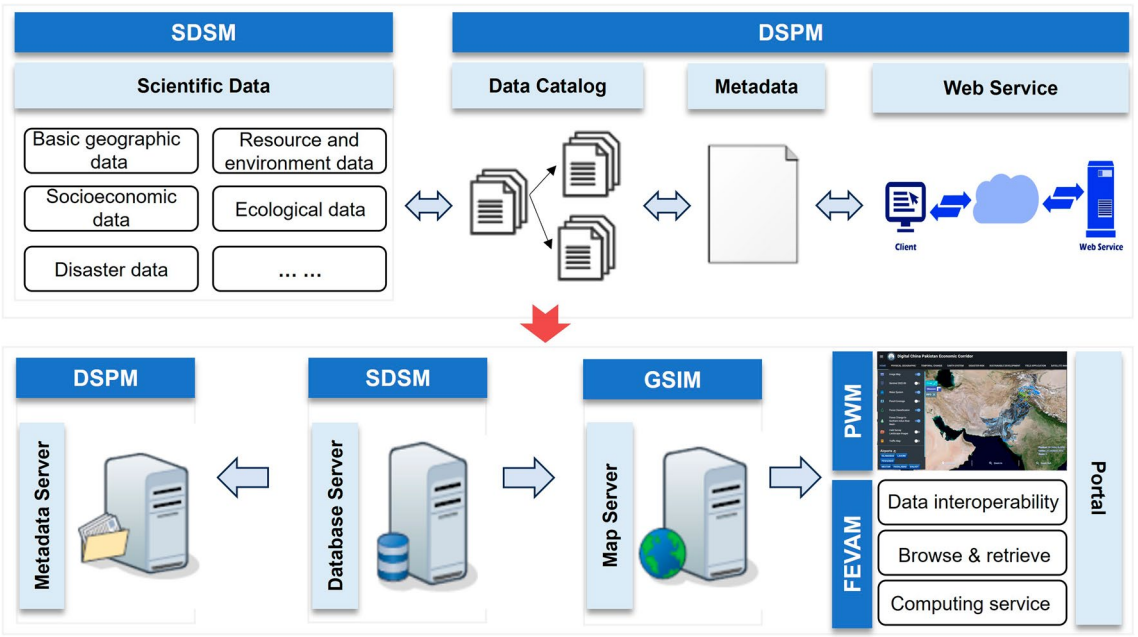


Fig. 1 Digital China-Pakistan Economic Corridor (CPEC) platform module design

Web Thematic Services (WTS). These services are seamlessly integrated into the portal, enabling interaction with the platform through standard HTTP (Hypertext Transfer Protocol) via web browsers. The platform's development focused on four key areas, including:

(1) Core Content Management System (CMS)

The CMS serves as a backend management tool that interacts with front-end applications through web service interfaces, handling user permissions and information publication. Developed with the Django framework (free and open-source, Python-based web framework) using Python and the open-source PostgreSQL for database management, the CMS provides functionalities for data storage, classification, retrieval, and downloading via web service application interfaces.

The CMS consists of two main components: an operational database for user permissions and service configurations, and a geographic information database for map data and metadata. To streamline disaster data processing, the CMS introduces innovations in processes such as data reorganization, database insertion, and data encoding. The reorganization process integrates key dataset information, including name, version, format, abstract, project affiliation, ownership, production unit, production time, update time, and other relevant metadata elements. For data insertion, the CMS uses standardized IDs for data encoding to manage datasets and ensure their relevance. Additionally, standardized metadata validation tools have been developed to ensure that datasets meet quality and compliance standards.

(2) Data Catalog Services.

In this catalog, the GeoJSON format (an open standardized format designed for representing simple geographical features) was adopted to enable cross-platform and cross-language data exchanges for spatial data. This study proposed an innovative metadata processing approach by integrating Pycsw (an Open Geospatial Consortium (OGC) catalog service for the web (CSW) server implementation written in Python) for metadata exchange and interoperability, while QGIS (GIS software) handles professional mapping. On the server side, Python combined with Geospatial Data Abstraction Library (GDAL/OGR) processes WMS data to extract key geographic details like layer names, types, and spatial extents. The data are then published following OGC standards and stored in spatial databases for efficient querying and analysis, enabling scalable and compliant geographic data applications.

(3) Interoperable Map Services System

The platform implements a map-tiling function for spatial data related to resources, environment, ecology, and disasters, and publishes WMS services based on client requests. To ensure data security, MapProxy software (an open-source proxy for geospatial data) handles the tiling process, while QGIS Server efficiently organizes and manages map deployment using a folder-naming system. Additionally, an automated task system updates statistical calculations and generates cached results, reducing response times for front-end users.

(4) Visualization System

An innovative front-end application was developed using the Vue3+Quasar framework (An incremental framework for building user interfaces), enabling integration with backend services to visualize maps and statistical charts. Echarts front-end library was used to graphically render statistical data, and open-source GIS leaflet tools were expanded with features like directory navigation, content exploration, attribute retrieval, and spatial measurement. Users can also draw geometrical elements such as points, lines, and polygons, enhancing interactivity with the visualized data.

2.2 Digital China-Pakistan Economic Corridor (CPEC) Platform Data Management

This section provides a detailed overview of the data management framework that strengthens the integrity, security, consistency, transparency, and accessibility of multisource and multidisciplinary data integrated into the platform.

2.2.1 Data Sources

The platform uses the shared data resources originating from the UNESCO International Knowledge Center for Engineering Sciences and Technology (IKCEST) Disaster Risk Reduction Knowledge Service System (Wang et al. 2020), and the CPEC data sharing platform for resources, environment, ecology, and disasters. Relevant regional research data are dynamically collected in the platform as well. Currently, the data shared on the platform do not include sensitive disaster data.

The data sources are categorized into eight major categories: physical geography, land cover, natural resources, ecological environment, natural hazards, remote sensing products, population, and socioeconomic factors. These were integrated into different thematic corridors by the framework.

2.2.2 Data Organization Framework

Disaster big data cover various aspects of the Earth's system. The innovative "Seven Corridors" framework (Fig. 2) addresses data integrity challenges through a hierarchical structure that organizes multisource and multidisciplinary data across critical domains. Each thematic corridor

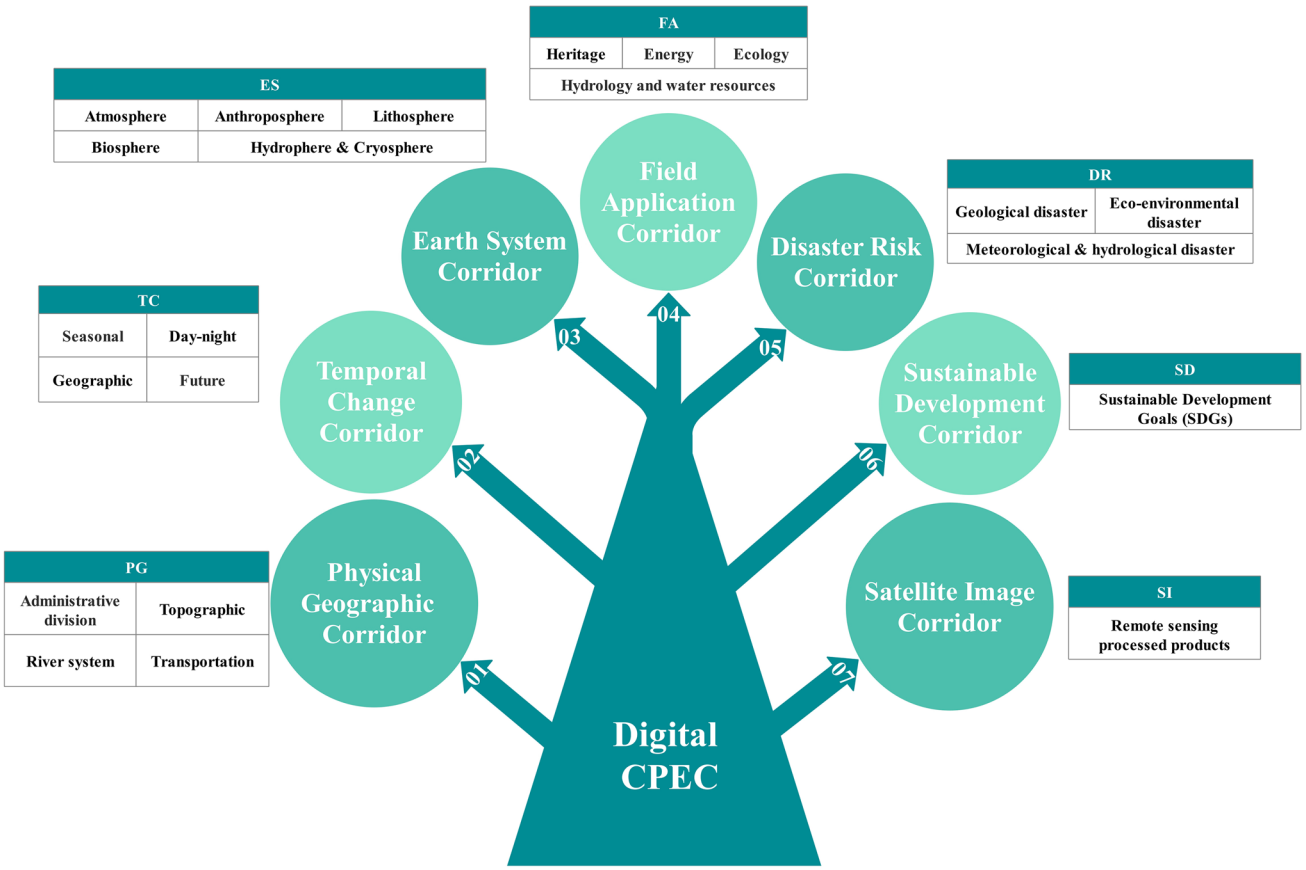


Fig. 2 Data system of the digital China-Pakistan Economic Corridor (CPEC) platform

comprises numerous secondary corridors. This organization ensures comprehensive, interdisciplinary data integration.

The "Seven Corridors" framework includes the following thematic corridors:

(1) Physical Geographic (PG) Corridor: Provides foundational data reflecting the natural environment of a region, with four sub-corridors—administrative division, topographic, river system, and transportation.
(2) Temporal Change (TC) Corridor: Captures spatiotemporal dynamics through four sub-corridors—seasonal, day-night, geographic, and future.
(3) Earth System (ES) Corridor: Integrates data from an Earth systems science perspective, consisting of five sub-corridors—atmosphere, biosphere, anthroposphere, hydrosphere & cryosphere, and lithosphere.
(4) Field Application (FA) Corridor: Targets specific application fields, with four sub-corridors—heritage, energy, ecology, and hydrology & water resources.
(5) Disaster Risk (DR) Corridor: Compiles different types of disaster-related data through three sub-corridors—geological disaster, eco-environmental disaster, and meteorological & hydrological disaster.
(6) Sustainable Development (SD) Corridor: Focuses on the United Nations SDGs and their indicators, generating assessment-ready data for the CPEC region.
(7) Satellite Image (SI) Corridor: Gathers and shares remote sensing processed products, including satellite and long-term archived historical image data.

2.2.3 Data Standards and Metadata

To ensure the reliability, quality, and cross-source alignment of the multisource and multidisciplinary data, we established standardized core metadata fields in the data integration process (Fig. 3). A unique set of metadata specifications with 18 key indicators (Table 1) and 38 metadata identifiers (Table 2) were designed to enhance metadata processing. This standardization simplifies the processes of data tracking, management, and citation, representing an innovative effort to systematically organize and manage diverse data types, thereby promoting more efficient data utilization and sharing. All datasets integrated into the Digital CPEC platform adhere to these metadata specifications. All datasets are findable, accessible, interoperable, and reusable in

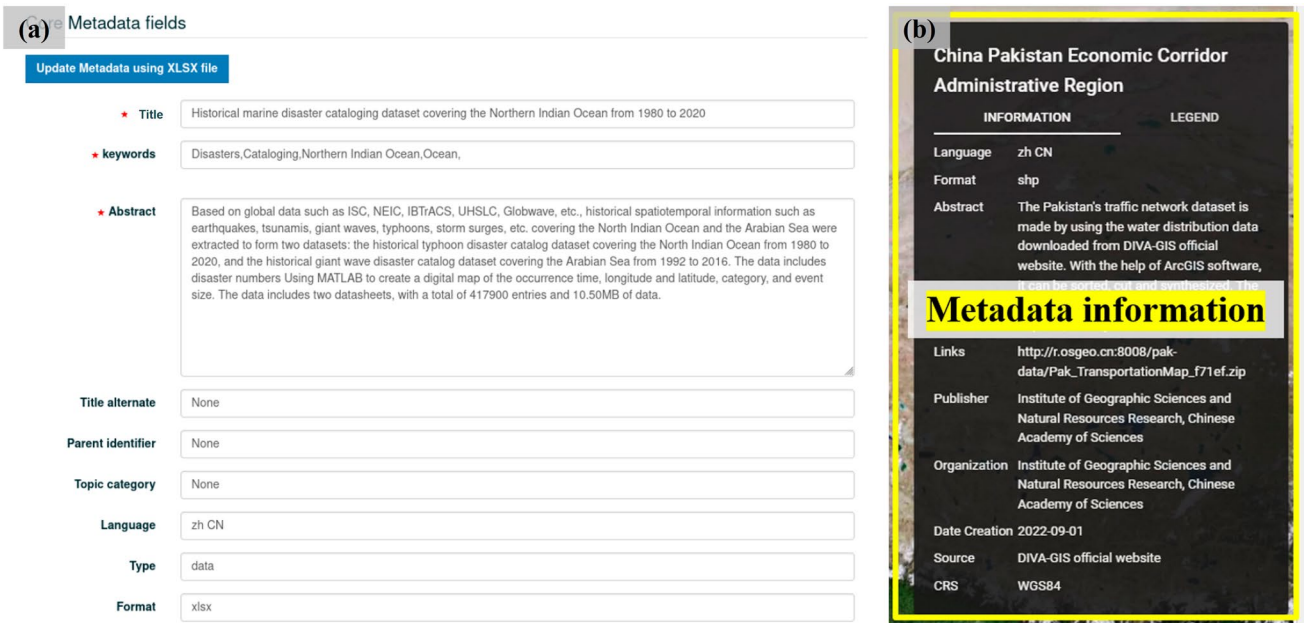


Fig. 3 Core metadata fields interface (a) in the data integration process and metadata information (b) shown in the Digital China-Pakistan Economic Corridor (CPEC) platform

Table 1 Metadata indicators

No.	Name	Description	No.	Name	Description
1	Title	Full Title	10	Source	Data Source
2	Title alternate	Short Alias	11	Links	Related Links
3	Parent identifier	Category	12	Time begin	Start Date
4	Topic category	Discipline Classification	13	Time end	End Date
5	Language	Language	14	Creator	Data Creator
6	Type	Data Type	15	Publisher	Data Publisher
7	Format	Data Format	16	Contributor	Data Contributor
8	Abstract	Full Text Description	17	Organization	Organization
9	Keywords	Keywords	18	Operator	Metadata Creator

accordance with the FAIR principles, supported by the open metadata on the platform (see Fig. 3).

2.2.4 Data Visualization Standards

To ensure consistency and uniformity in data visualization, the Digital CPEC employs standardized mapping templates and specifications. The WGS 1984 coordinate system, combined with the UTM projection system, provides a suitable map projection for specific regions. Pakistan spans UTM Zones 41 to 43, with Zone 42N (longitude range approximately 66°E to 72°E) covering the central region of the country. Therefore, the Digital CPEC platform adopts the WGS 1984 UTM Zone 42N projection coordinate system for all spatial data. The platform also standardizes curved grids, uses the Times New Roman font, applies fixed scale sizes, and ensures consistent placement of compasses and legends. These measures enhance map readability and maintain uniformity, contributing to the clarity and reliability of visualized data.

3 Results

This section presents the Digital CPEC platform's core interface architecture and functionality, followed by a detailed exposition of its data management specifications.

Table 2 Metadata identifiers in the schema

No.	Name	Description	No.	Name	Description
1	Identifier	Identifier: Auto-generated	20	Wkt geometry	Well-known Text of Geometry
2	Classification	Data Classification	21	Service type	Service Type
3	Md source	Origin of Resource	22	Service type version	Service Type Version
4	Resource language	Source Language	23	Operation	Allowed Operation
5	Keywords type	Keywords Type	24	Coupling type	Coupled Resource
6	Relation	Related Description	25	Operator identifier	Data Administrator Identifier
7	Security constraints	Security Constraints	26	Operator name	Data Administrator Name
8	Access constraints	Access Constraints Related to the Service	27	Degree	Hierarchy Level
9	Other constraints	Other Restrictions	28	Lineage	Lineage
10	Date	Dates	29	Responsible party role	Responsible Organization
11	Date revision	Version Date	30	Specification title	Specification Name
12	Date creation	Creation Date	31	Specification date	Specification Date
13	Date publication	Publication Date	32	Specification date type	Specification Date Type
14	Date modified	Revision Date	33	Insert date	Insert Date
15	CRS	Coordinate Reference System	34	XML	Complete XML Document
16	Geodescode	Geographic Description Code	35	Any text	Complete XML Representation
17	Denominator	Scale Denominators in Spatial Resolution	36	Type name	Metadata Type Name
18	Distance value	Spatial Extent	37	Schema	Metadata Schema
19	Distanceuom	Unit of Measure (UoM) of the Distance	38	Condition applying to access	Terms and Conditions for Access and Use

For more information, please refer to: ISO 19115-1:2014 Geographic information—Metadata Part 1: Fundamentals

3.1 Digital China-Pakistan Economic Corridor (CPEC) Platform

The Digital CPEC platform[1] developed in this study is an innovative and integrated geographic information service system that showcases geographic information as well as represents significant advances in service effectiveness in a fully digitalized view. It offers efficient, user-friendly access to geographic information, supporting interdisciplinary research and decision making within the CPEC region. By integrating functionalities such as online map browsing, data directory navigation, and interactive data exploration, the platform enhances both data accessibility and utilization efficiency, optimizing the overall user experience. The data integrated into the platform have been widely accessed since June 2017, with over 1.324 million downloads as of December 2024. This highlights the significant potential for further integrating reanalyzed data into the platform, in alignment with the data-sharing standards established in this study.

The Digital CPEC platform currently offers users the following six key features (Fig. 4): (1) Display of statistical tables for key information parameters; (2) Visualization of raster data, with a focus on dynamic changes within the TC corridor, allowing users to observe the temporal variations of parameters; (3) Key area positioning (for example, airports, harbors); (4) Measurement tools, supporting irregular area and distance measurements; (5) Vector drawing tools, supporting drawing features such as polyline, polygon, circle, and point; (6) Instant real-world scenarios, displaying real-time views of specific locations at a given moment. Users can click on the corresponding buttons to use the relevant functions.

3.2 Data Management Specifications

In the context of the data-intensive study paradigm, open data management has become an integral part of national strategies, with many countries, including China, incorporating it into their development plans (Boulton 2016). To enhance data management within the CPEC, this study established four key standards designed to ensure efficient, organized, and secure data management and sharing on the Digital CPEC platform.

(1) Digital CPEC Platform Data Management Standards: These standards outline the comprehensive lifecycle management of scientific data on the platform. It details the required protocols for data collection, organization,

[1] http://cpjrc-quasar.gislab.cn/

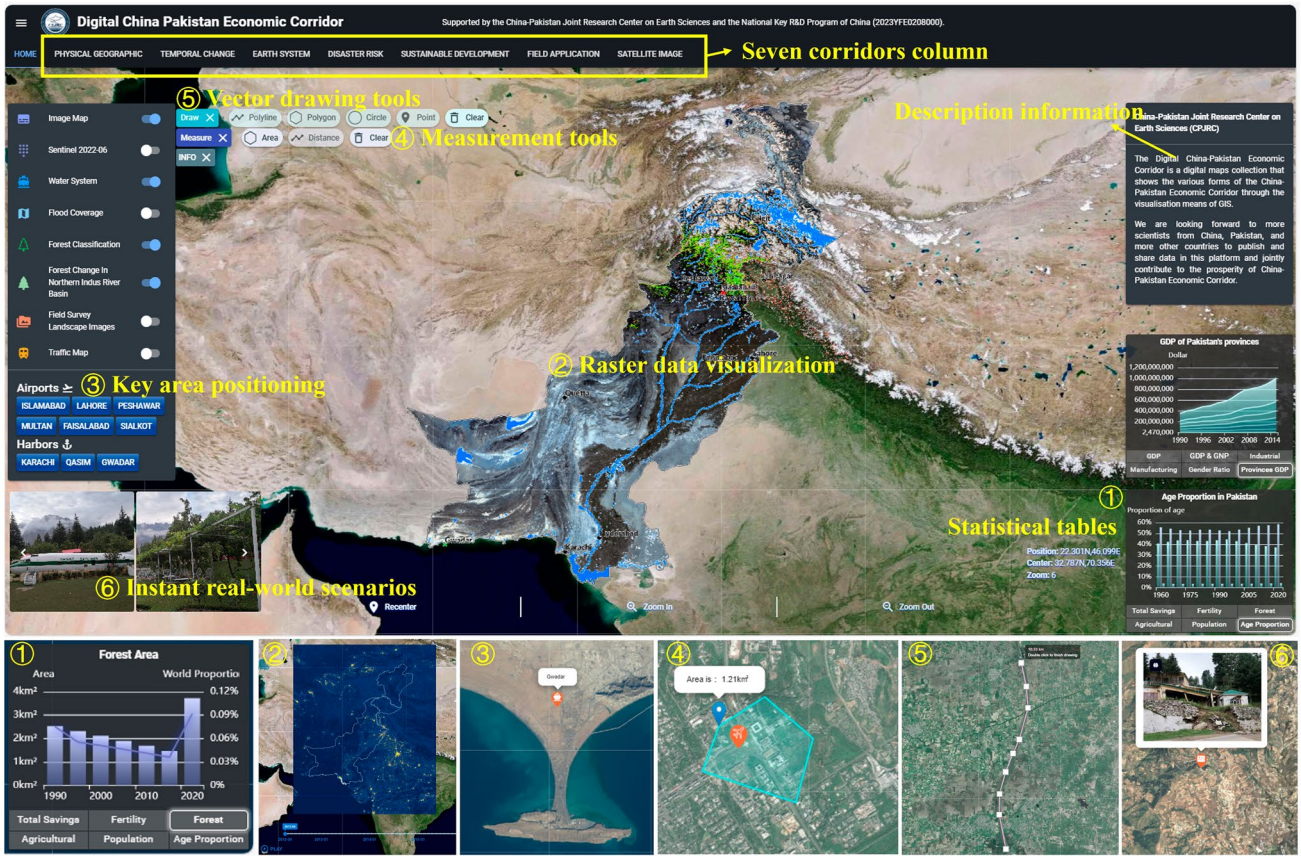


Fig. 4 Homepage interface of the digital China-Pakistan economic corridor (CPEC) platform. Items ①–⑥ refer to six key features along with corresponding examples

archival prepublication, and ongoing data management and backup procedures, ensuring that all data adhere to high standards of integrity and accessibility.

(2) Digital CPEC Platform Metadata Standards: This standard establishes guidelines for system metadata, which is essential for the effective cataloging and management of disaster data. By enhancing data retrieval, exchange, integration, and services, this standard promotes interoperability and usability across various research domains.

(3) Disaster Science Data Documentation Specification: A disaster science data document template is developed specifically for data introduction, which includes critical content details such as disciplinary scope, storage, quality control, sharing methods, and intellectual property rights statements.

(4) Digital CPEC Platform Sharing Service Specification: This specification governs the platform's data-sharing services, emphasizing user-centric needs. It addresses various aspects of access, including open access policies, service classification, content focus, and legal rights protection.

4 Practical Applications

This section highlights three key applications to showcase that the Digital CPEC platform holds significant potential in advancing disaster resilience and strategy insights throughout the entire lifecycle of disaster management. All original data and reanalyzed data were reintegrated within the platform, in accordance with the data management specifications (Sect. 3.2) and metadata specifications (Sect. 2.2.3), and can be reused for the public following the FAIR principles.

4.1 Digital China-Pakistan Economic Corridor (CPEC) Platform Supports Forest Ecological Change and Driving Factor Analyses in the Upper Indus River Basin

The upper Indus River Basin is a critical yet fragile water tower for the Indian subcontinent, supplying water to 230 million people. Clarifying ecological environment variations, especially forest changes and driving factors, are essential for the sustainable development of this region. Based on the multisource data integrated into the platform, including Landsat (SI Corridor) and meteorological reanalysis data (TC Corridor), and using methods such as the LandTrendr algorithm, decision tree classification models, InVEST assessment models, and GIS statistical analysis tools, we analyzed high-resolution spatiotemporal information on forest changes and driving factors in the basin from 1990 to 2020.

The study revealed that the general trends of disturbance and recovery appear significantly different between regions, with Nepal showing the highest forest stability, India showing the most disturbances, and Pakistan and China showing the most significant recoveries. Forest degradation was the main cause of disturbances, followed by agricultural conversion. Natural recovery was the primary driver of forest restoration, followed by recovery from human interventions. These findings have been published in Yan and Wang (2022) and Yan et al. (2022), and shared on the FA Corridor modules (Fig. 5).

These findings can support the development of more effective forest management policies at the regional and national scales. By identifying the dominant drivers of forest change and regional variations in recovery and disturbance dynamics, policymakers can design targeted interventions, such as prioritizing afforestation efforts in highly disturbed regions like India, strengthening conservation policies in Nepal to maintain its high forest stability, and enhancing reforestation programs in Pakistan and China to sustain their recovery momentum. Additionally, the integration of multisource data enables policymakers to assess the effectiveness of existing policies and refine strategies based on observed trends, ensuring adaptive and region-specific forest management approaches.

4.2 Digital China-Pakistan Economic Corridor (CPEC) Platform Supports Rapid Flood Relief in the Southern Plains of Pakistan

In mid-June 2022, Pakistan experienced its highest rainfall in approximately 30 years. To respond to this unprecedented flood, an emergency data module was rapidly established based on the platform's scalable functionality, integrating available data according to data management standards, especially the metadata specifications to ensure data quality. This supported our monitoring and analysis of the flood. For example, using Sentinel-1 satellite imagery integrated in the SI Corridor and VV and VH polarization methods, flood extents before, during, and after the disaster were extracted. Spatial consistency

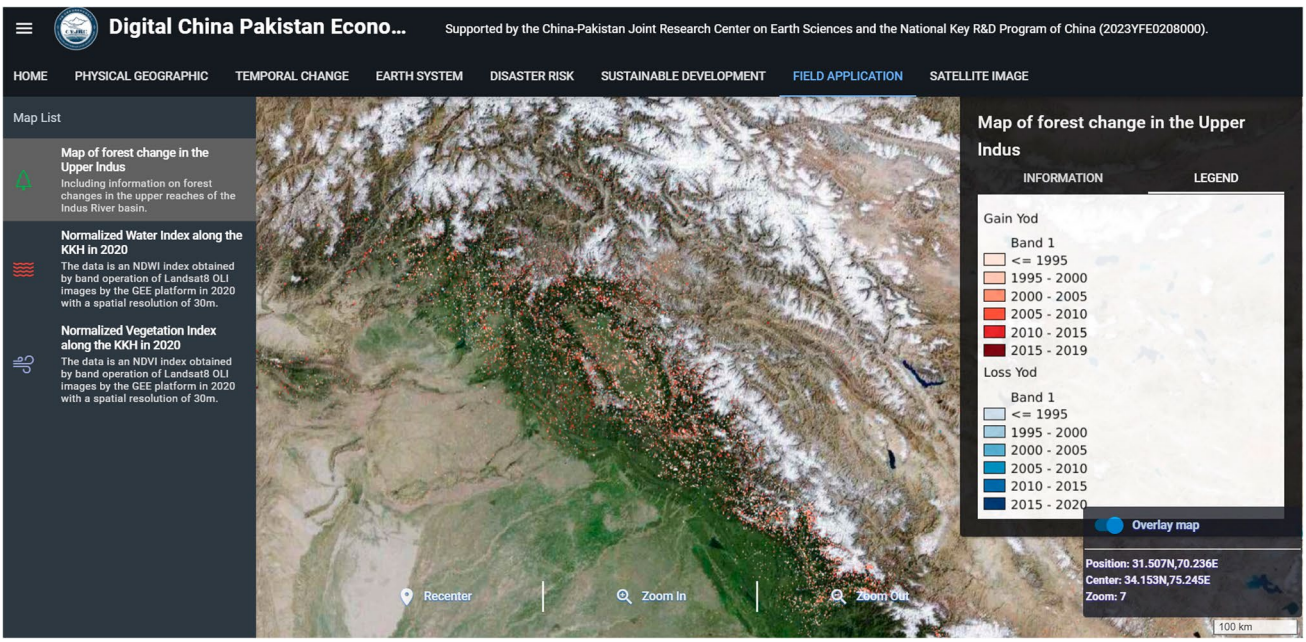


Fig. 5 Forest change in the upper Indus

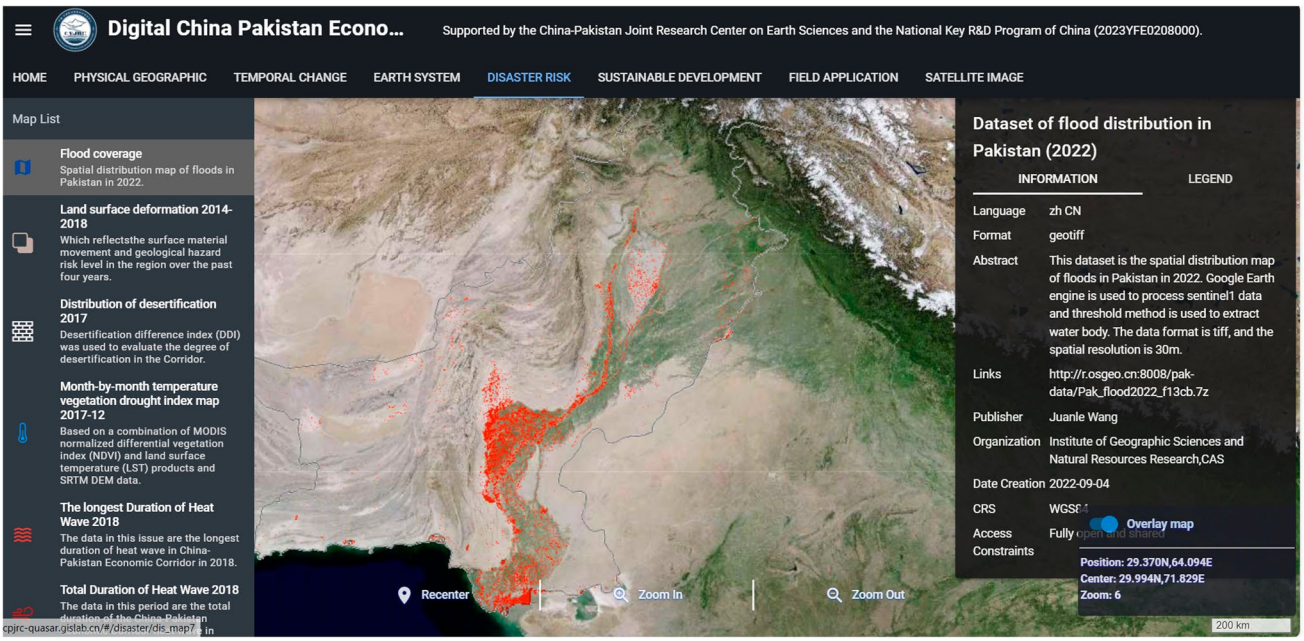


Fig. 6 Flood coverage of Pakistan in 2022

analysis was then conducted using the flood distribution data, quantifying the impact of the flood on buildings, roads, and farmland nationwide. This result was published in Wang et al. (2024). Additionally, a blue-spot model was developed, classifying the risk level of dam-break floods into five categories—extremely low, low, medium, high, and extremely high—based on DEM data integrated in the PG Corridor. A dam-break flood risk distribution map within a 100 km radius of the dam was generated. The relevant flood datasets and risk maps (Fig. 6) generated during this process were released in the DR Corridor.

These results directly inform policy making by providing a data-driven foundation for emergency DRR strategies. The flood extent and impact assessments enable authorities to prioritize emergency response and allocate resources more effectively. The dam-break flood risk classification supports the formulation of stricter infrastructure safety regulations and improved early warning systems. The increased accessibility of standardized datasets fosters transparency in disaster management, facilitating cross-border collaboration in flood response and mitigation planning. Furthermore, the adoption of FAIR principles has not only enhanced data discoverability and reusability, strengthening long-term policy frameworks for climate resilience and disaster preparedness, but also significantly increased data access and utilization, with the shared datasets receiving 3,148 visits and 282.45 GB of downloads within five months of their release.

4.3 Digital China-Pakistan Economic Corridor (CPEC) Platform Supports Ecological and Environmental Assessment in the Lower Indus River Basin

The lower Indus River Basin is pivotal for advancing the United Nations SDGs 14 and 15, serves as Pakistan's "granary" and hosts the world's sixth largest mangrove ecosystem. Clarifying its ecological environment health is essential for ecological and environmental protection, increasing the sustainability of this region.

Based on the Landsat 8 time-series data integrated into the platform (SI corridor), we developed a deep learning model, MSNet, to generate land cover maps of the lower Indus River Basin region before and after flooding in 2022. Using these maps, we then evaluated the landscape ecological health of the region with the VOR model. The results indicate beneficial effects of floods and the improved ecological state of the area, with a decrease in "pathological" areas and an increase in very healthy, healthy, and sub-healthy zones. These findings have been published in Xu et al. (2023) and are shared in the SD Corridor (Fig. 7).

These results provide critical insights for policy making, emphasizing the need for policies that recognize the ecological benefits of natural flooding events. First, land use planning policies should prioritize adaptive ecological restoration in post-flood landscapes, promoting sustainable land management practices. Second, the results highlight

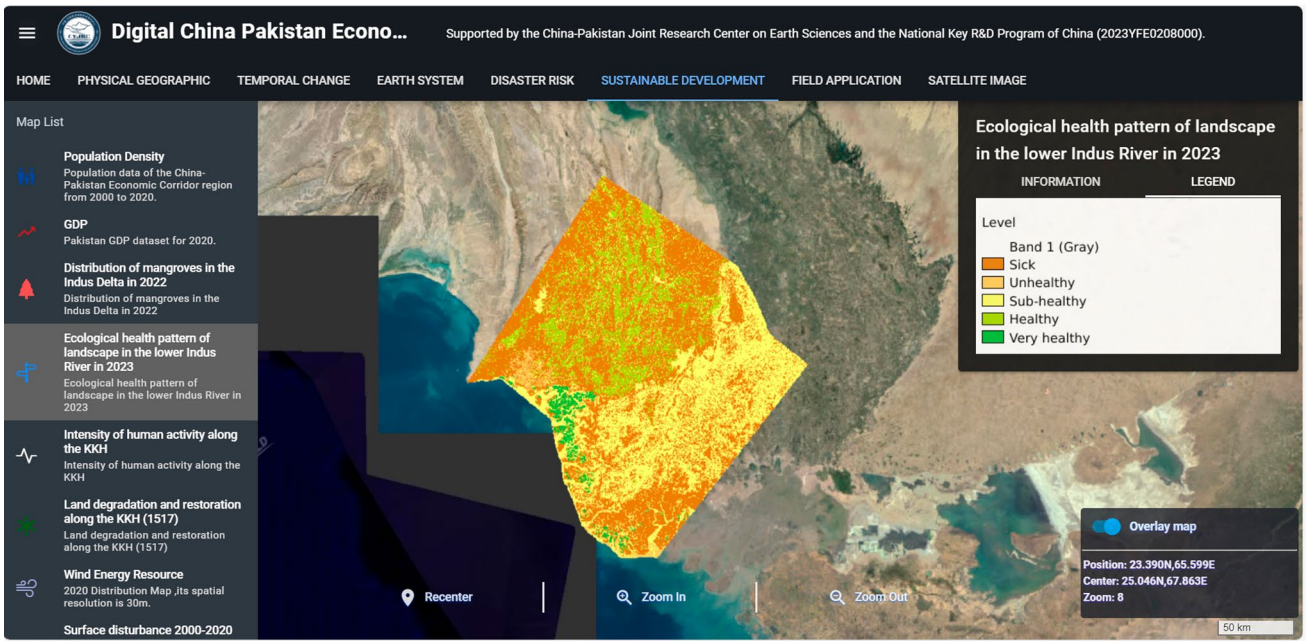


Fig. 7 Ecological health pattern of landscape in the lower Indus River Basin in 2023

the necessity of integrating real-time ecological monitoring into disaster response frameworks to inform evidence-based policy decisions.

5 Discussion

This section focuses on discussing the novelty, contributions, and limitations of the Digital CPEC platform.

5.1 Novelty of the Digital China-Pakistan Economic Corridor (CPEC) Platform Compared with Other Disaster Data Platforms

The Digital CPEC platform introduces several key innovations that address gaps in existing DRM systems like EM-DAT and Copernicus EMS. Unlike these systems, which often rely on centralized, static databases, the Digital CPEC platform provides a visualization capability for all the digital resources, and all data can be shown on a map in CPEC regions. Second, Digital CPEC integrates real-time data from diverse sources—such as remote sensing and socio-economic data—using a modular and scalable design. This approach enhances flexibility, enabling the platform to adapt to evolving user needs and technological advancements. The integration and sharing of reanalysis data and results obtained from the research of this region reduce data fragmentation. The development of personalized modular tools enhances user interoperability, offering a more tailored and efficient user experience.

5.2 Contributions to National and International Disaster Risk Management Frameworks

The design of the Digital CPEC platform is aligned with global and national DRM frameworks, ensuring its scientific rigor and effectiveness in addressing complex disaster challenges. At the international level, the platform responds to the core objectives of the Sendai Framework for Disaster Risk Reduction 2015–2030 by enhancing data accessibility and transparency, thereby supporting data-driven DRM, particularly in risk assessment and data integration. The platform is highly aligned with the United Nations Sustainable Development Goals and directly contributes to the 2030 Agenda for Sustainable Development. Through open data sharing, scientific knowledge dissemination, and multi-stakeholder collaboration, the platform accelerates the localized and regional implementation of the SDGs.

At the national level, the Digital CPEC platform serves as a technical support tool for DRM, complementing Pakistan's National Disaster Management Plan and National Disaster Risk Reduction Policy. By integrating real-time scientific data and reanalysis datasets, the platform optimizes the full-cycle management of disaster data, enhancing risk identification and emergency response capabilities. Furthermore, its open science architecture lowers technological barriers, enabling local decision makers to more efficiently access, analyze, and use disaster risk information, directly incorporating it into emergency strategies and policy formulation.

5.3 Contributions to Disaster Risk Reduction and Management of Pakistan

Recent studies emphasize that technology and technological innovation (Latief et al. 2024; Waheed, Kousar, et al. 2025), as well as data management systems for data transparency and access to reliable information (Waheed et al. 2024), are key strategies for global cooperation in addressing climate change and advancing environmental sustainability. Digitalization is the process of integrating digital technologies to strengthen connections among people, systems, companies, products, and services (Ritter and Pedersen, 2020), thereby improving transparency and traceability within systems (Antikainen et al. 2018). The Digital CPEC platform exemplifies how digitalization integrates advanced technologies to address complex global challenges, introducing innovative solutions to the digital divide, data integration, and disaster management in the CPEC region.

5.3.1 Mitigating Internal Weaknesses: Bridging the Digital Divide and Promoting International Cooperation

The Digital CPEC platform is centered on an open data policy, multilingual interface, and open-source framework, which reduce technological and economic barriers while integrating transnational data resources into a unified system. The use of open-source technologies, such as Python, PostgreSQL, and QGIS, ensures cost-efficiency and accessibility, making the platform particularly suitable for developing regions. These tools support the platform's scalability and interoperability, allowing it to integrate data from various stakeholders and provide seamless services. The platform facilitates data sharing through the open-source framework aligned with the UNESCO 2021 Open Science Recommendation, enhancing data transparency, knowledge exchange, and research impact.

5.3.2 Addressing External Threats: Innovative Multidimensional Data Integration Framework

The platform's adoption of the FAIR principles ensures that data are findable, accessible, interoperable, and reusable, promoting greater transparency, scalability, and collaboration in disaster management. This novel approach significantly improves data sharing, transparency, and real-time decision making in comparison to traditional systems, driving multidimensional innovation model encompassing policy, data, software, and deployment modules.

(1) Policy Assurance: The platform adheres to international cooperation policies and the data security-sharing policies, laws, and regulations of China and Pakistan. It has established the "Digital CPEC Platform Data Management Standards" and the "Digital CPEC Platform Sharing Service Specification," providing institutional guarantees for the security, privacy, and sustainable development of cross-border data transfer and processing, as well as for global trust in the platform.

(2) Data Integration: The platform integrates transnational data resources across the CPEC region and builds a data management system covering seven major corridors to ensure data integrity. By designing 18 key metadata indicators and 38 metadata identifiers, and applying a unified data management and processing approach (as outlined in the "Digital CPEC Platform Metadata Standards" and "Disaster Science Data Documentation Specification"), the platform ensures data quality and digital consistency, reduces data fragmentation, and improves data utilization efficiency.

(3) Software Support: Based on the open science framework, the platform provides a multilingual, internationalized user interface to promote global data sharing and exchange, thereby deepening international cooperation.

(4) Custom Deployment: Customized functions are developed based on the specific needs of different countries, industries, and organizations, offering targeted value-added services (for example, tools for area and length measurement) to achieve efficient visualization and precise delivery of disaster information, thereby enhancing user satisfaction and platform value.

5.3.3 From Data Sharing to Deep Applications

The fully digital platform drives innovation from data sharing to in-depth applications, primarily manifested in two dimensions.

(1) Support for Data Analysis: For example, the nighttime light data from the TC Corridor were used to monitor human activities in the coastal areas of Balochistan Province (Chaudhry et al. 2024); The ES Corridor supports research on natural hazard-related disaster mechanisms, such as the impact of glacier melting and upstream water resource protection on downstream flooding (Yan et al. 2022; Wang et al. 2024). The SD Corridor provides data related to the SDGs, including agriculture (SDG2), water resources (SDG6), renewable energy (SDG7), urbanization (SDG11), and climate change (SDG13) (Yan and Wang 2022; Yan et al. 2022; Xu et al. 2023).

(2) Support for Decision Making: The platform facilitates decision making through two core mechanisms: (1) Precision Decision Support—Enhanced by personalized modular tools that improve decision accuracy, exemplified by real-time scenario photo sharing and timely disaster response data coordination, effectively

reducing critical information response delays; (2) Evidence-Based Policy Optimization—By integrating regional research outputs including reanalysis datasets, the platform achieves full lifecycle management of disaster data while providing scientific support for resilience policy formulation, environmental sensitivity assessments, and disaster management policy effectiveness evaluation. This mechanism effectively addresses Pakistan's systemic gaps in environmental policy assessment (particularly post-implementation analysis) (Huang et al. 2017).

5.4 Disadvantages and Future Research Directions

Artificial intelligence (AI) and standardized digital processes are key to driving digital transformation (Nadeem et al. 2024). With the rapid development of digital technologies and data resources, the application and innovation of AI, machine learning, deep learning, large-scale modeling, and digital twin technologies in disaster management have significantly improved the accuracy of disaster forecasting, making them indispensable key technologies in disaster management (IPCC 2023; Roberts et al. 2023; Nadeem et al. 2024). These advancements provide new perspectives for disaster risk assessment and response strategies, particularly within the context of Digital Pakistan initiatives, where they will profoundly change disaster management practices.

However, Challenges persist in the process of standardized digitization, such as limited international data providers, difficulties in multisource data alignment, and cross border data security issues. Currently, the majority of the data on the platform is still provided by the research team, while user-contributed data, particularly from international users, remains scarce. This creates an uneven data landscape that led to incomplete or skewed data representation. In addition, the current Digital CPEC platform addresses data alignment by using standardized metadata and fixed projection methods. Differences in data formats and standards across sources remain a significant technical barrier. This requires further optimization of data management processes to ensure smooth integration of data from various sources. For data security, the data shared on the platform do not include sensitive disaster data currently. In the future, the platform will further enhance the CPEC data system. In addition to involving more data contributors, it will develop specialized data products, particularly tiered products, to provide targeted services for different stakeholders. This will expand the scope of data sharing and application while ensuring the security of cross-border data sharing. Furthermore, the platform will optimize data management standards and establish classification and tiered management systems for different users to advance the global process of data standardization.

Given the significant potential of sustainability strategies and technological innovation in addressing climate change and advancing environmental sustainability, future research on platform data should focus on enhancing the innovative applications of AI and machine learning throughout the DRM lifecycle. This includes disaster prediction modeling, optimizing resource allocation, and improving real-time decision making for disaster response, providing knowledge-based services to enhance disaster management efficiency and accuracy. Furthermore, green industries such as sustainable tourism have been proven to contribute to sustainable development (Latief et al. 2024). Therefore, future research should further explore the deep integration of sustainability strategies across various industries, promoting data-driven adaptive development strategies and providing scientific support for policy formulation and industry optimization. These innovative applications will comprehensively enhance the platform's decision-making support capabilities, foster cross-sector collaboration, and drive data-driven intelligent management and sustainable development practices.

Nonetheless, while the platform offers technological and data integration innovations, it does not sufficiently engage with the issue of policy alignment across governance sectors. In the context of complex, multi-stakeholder initiatives like CPEC, the lack of coherent policy frameworks can undermine the effective implementation of disaster management and resilience strategies. Fragmented governance—with environmental policies, sectoral development plans, and disaster risk management efforts working in silos—is a major risk to the successful operation and long-term sustainability of digital platforms, limiting the platforms' potential to underpin change in disaster prevention and mitigation. Building on the insights of Waheed, Khan, et al. (2025), who emphasized the need for cross-sectoral coordination mechanisms to harmonize CPEC plans with national and regional governance frameworks, future research should go beyond technical enhancement to investigate how digital platforms can actively support institutional coherence. This includes exploring their potential role in bridging policy divides, aligning disaster management objectives with environmental and infrastructure development goals, and fostering multi-level governance collaboration.

6 Conclusion

This article presented the development of the Digital CPEC platform, a digital solution designed to address disaster management challenges in the CPEC region, while offering transferable insights for broader global contexts. This platform fosters innovation by addressing both internal weaknesses and external challenges in Pakistan's digital initiatives. It bridges the digital divide and establishes a

multidimensional data integration framework, enhancing transparency, accessibility, and interoperability, thereby setting a new benchmark for advancing disaster management capacities in underdeveloped and complex, multi-stakeholder regions. The key contributions are as follows.

First, the platform's open-source architecture, built on open science standards and technologies, significantly reduces technical and economic barriers, providing a scalable and cost-effective model applicable in disaster-prone regions worldwide. Unlike traditional static repositories, the platform introduces grid-based mapping for dynamic dataset visualization and a fully digital approach that enhances clarity and accuracy. Tailored analytical tools further improve user interaction, making the platform accessible to diverse users and fostering cross-sector collaboration.

Second, guided by open science principles, this study established a standardized, innovation-driven digital process that spans the entire data lifecycle, offered a structured approach to consolidate fragmented data across sectors, showcasing how comprehensive frameworks can underpin large-scale disaster management systems. Specifically, the development of a seven-corridor data structure ensures comprehensive data coverage and effective integration, while four standardized frameworks and metadata storage standards maintain data security and consistency. These standards enable seamless alignment and efficient sharing of multi-source data, addressing common challenges in both national and cross-border disaster data management.

Third, the platform offers strategic insights into enhancing environmental sustainability, demonstrating how digital platforms can empower the alignment of disaster resilience goals with sustainable development objectives. For example, forest ecosystem driver analysis aids in designing targeted interventions; rapid flood extent and impact assessments help prioritize emergency responses and optimize resource allocation; and post-disaster evaluations highlight the ecological value of natural flooding, supporting policies that balance risk reduction with environmental sustainability. This study fully demonstrates the positive contribution of open science to DRR and the promotion of international cooperation through digital technologies.

Acknowledgments This work was supported by the National Key R&D Program of China (Grant No. 2023YFE0208000), the China-Pakistan Joint Research Center on Earth Sciences, and the Construction Project of China Knowledge Centre for Engineering Sciences and Technology (Grant No. CKCEST-2023-1-5).

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Ahmed, R. 2019. Ecommerce in Pakistan: Challenges & opportunities. *Asian Journal of Economics, Finance and Management* 2(4): 131–139.

Antikainen, M., T. Uusitalo, and P. Kivikytö-Reponen. 2018. Digitalisation as an enabler of circular economy. *Procedia Cirp* 73: 45–49.

Atta Ur, R., and R. Shaw. 2015. Introduction and approaches of disaster risk reduction in Pakistan. In *Disaster risk reduction approaches in Pakistan*, ed. R. Shaw, 3–29. Tokyo: Springer.

Bhattacharjee, D. 2015. China Pakistan Economic Corridor. https://ssrn.com/abstract=2608927. Accessed 22 May 2015.

Boulton, G. 2016. International accord on open data. *Nature* 530(7590): 281–281.

Butt, H.D., M.U. Aijaz, M.A. Shamim, K.S. Lodhi, A. Hayat, and M. Mazhar. 2024. Leveraging CPEC for a thriving blue economy and coastal development. *Migration Letters* 21(S8): 1285–1321.

Chaudhry, T.T., M. Asif, and M.M. Hussain. 2024. Using nighttime light data to estimate sub-national GDP and growth in Pakistan. https://doi.org/10.2139/ssrn.4814807. Accessed 6 May 2024

Fischer-Preßler, D., D. Bonaretti, and D. Bunker. 2024. Digital transformation in disaster management: A literature review. *The Journal of Strategic Information Systems* 33(4): Article 101865

Fujita, S., and M. Hatayama. 2023. Estimation method for roof-damaged buildings from aero-photo images during earthquakes using deep learning. *Information Systems Frontiers* 25(1): 351–363.

GOP (Government of Pakistan). 2013. Pakistan: National disaster risk reduction policy (2013). Islamabad: Government of Pakistan, Ministry of Climate Change. https://www.refworld.org/legal/decrees/natlegbod/2013/en/121377. Accessed 6 Jan 2025.

GOP (Government of Pakistan). 2017. National sustainable development strategy 2017. Islamabad: Ministry of Climate Change. http://www.mocc.gov.pk. Accessed 9 Dec 2021.

Huang, Y., T.B. Fischer, and H. Xu. 2017. The stakeholder analysis for SEA of Chinese foreign direct investment: The case of "One Belt, One Road" initiative in Pakistan. *Impact Assessment and Project Appraisal* 35(2): 158–171.

IPCC (Intergovernmental Panel on Climate Change). 2023. *Climate change 2021 – The physical science basis*. Cambridge: Cambridge University Press.

Khan, A., S. Gupta, and S.K. Gupta. 2020. Multi-hazard disaster studies: Monitoring, detection, recovery, and management, based on emerging technologies and optimal techniques. *International Journal of Disaster Risk Reduction* 47: Article 101642

Latief, R., Y. Pei, S. Ahmad Javeed, and U. Sattar. 2024. Environmental sustainability in the Belt and Road Initiative (BRI) countries: The role of sustainable tourism, renewable energy and technological innovation. *Ecological Indicators* 162: Article 112011

Mekonnen, M.M., and A.Y. Hoekstra. 2016. Four billion people facing severe water scarcity. *Science Advances* 2(2): Article e1500323

Mondal, S.K., J. Huang, Y. Wang, B. Su, J. Zhai, H. Tao, G. Wang, T. Fischer, et al. 2021. Doubling of the population exposed to drought over South Asia: CMIP6 multi-model-based analysis. *Science of the Total Environment* 771: Article 145186

Mueller, V., C. Gray, and K. Kosec. 2014. Heat stress increases long-term human migration in rural Pakistan. *Nature Climate Change* 4(3): 182–185.

Nadeem, M., Y. Ali, O. Rehman, and L.T. Saarinen. 2024. Barriers and strategies for digitalisation of economy in developing countries: Pakistan, a case in point. *Journal of the Knowledge Economy* 15(1): 4730–4749.

Nizam, H.A., K. Zaman, K.B. Khan, R. Batool, M.A. Khurshid, A.M. Shoukry, M.A. Sharkawy, and F. Aldeek et al. 2020. Achieving environmental sustainability through information technology: "Digital Pakistan" initiative for green development. *Environmental Science and Pollution Research* 27(9): 10011–10026.

Otto, F.E.L., M. Zachariah, F. Saeed, A. Siddiqi, S. Kamil, H. Mushtaq, T. Arulalan, K. AchutaRao, et al. 2023. Climate change increased extreme monsoon rainfall, flooding highly vulnerable communities in Pakistan. *Environmental Research: Climate* 2(2): Article 025001

Phong, T., S. Sonak, and R. Shaw. 2009. Disaster, environment and development: Opportunities for integration in Asia-Pacific region Global Challenges and Local Solutions. *Disaster Management* 25: 400–423.

Ritter, T., and C.L. Pedersen. 2020. Digitization capability and the digitalization of business models in business-to-business firms: Past, present, and future. *Industrial Marketing Management* 86: 180–190.

Roberts, J., T. Lüddecke, S. Das, K. Han, and S. Albanie. 2023. GPT-4GEO: How a language model sees the world's geography. *arXiv*: 2306.00020.

Shahzad, A. 2022. Pakistan's Sharif lands in Beijing to discuss economy, debt. Islamabad: Reuters. https://www.reuters.com/world/asia-pacific/pakistans-sharif-lands-beijing-discuss-economy-debt-2022-11-01/. Accessed 2 Nov 2022.

Song, T., H. Liu, and Y. Wuzhati. 2023. Decentralization and collaborative disaster governance for the China-Pakistan Economic Corridor: Current trends and implications. *Journal of Resources and Ecology* 14(5): 974–982.

Su, B., J. Huang, M. Gemmer, D. Jian, H. Tao, T. Jiang, and C. Zhao. 2016. Statistical downscaling of CMIP5 multi-model ensemble for projected changes of climate in the Indus River Basin. *Atmospheric Research* 178: 138–149.

Waheed, A., T. Bernward Fischer, and M.I. Khan. 2021. Climate change policy coherence across policies, plans, and strategies in Pakistan – Implications for the China-Pakistan Economic Corridor Plan. *Environmental Management* 67(5): 793–810.

Waheed, A., T.B. Fischer, S. Kousar, and M.I. Khan. 2023. Disaster management and environmental policy integration in Pakistan – An evaluation with particular reference to the China-Pakistan Economic Corridor Plan. *Environmental Science and Pollution Research* 30(48): 105700–105731.

Waheed, A., M.I. Khan, and T.B. Fischer. 2025. Assessing environmental policy coherence in Pakistan: Implications for the sustainability of Belt and Road Initiative's China-Pakistan Economic Corridor Plan (2017–2030). *Journal of Environmental Policy & Planning*. https://doi.org/10.1080/1523908X.2025.2490265.

Waheed, A., S. Kousar, M.I. Khan, and T.B. Fischer. 2024. Environmental governance in Pakistan: Perspectives and implications for the China-Pakistan Economic Corridor Plan. *Environmental and Sustainability Indicators* 23: Article 10044

Waheed, A., S. Kousar, M.I. Khan, and T.B. Fischer. 2025. Expert perceptions on CPEC sustainability appraisal: SWOT analysis for informed policy decisions. *Environmental Science Policy* 164: Article 104009

Wang, J., K. Bu, F. Yang, Y. Yuan, Y. Wang, X. Han, and H. Wei. 2020. Disaster risk reduction knowledge service: A paradigm shift from disaster data towards knowledge services. *Pure and Applied Geophysics* 177(1): 135–148.

Wang, J., K. Li, L. Hao, C. Xu, J. Liu, Z. Qu, X. Yan, and M.M. Sajjad et al. 2024. Disaster mapping and assessment of Pakistan's 2022 mega-flood based on multi-source data-driven approach. *Natural Hazards* 120(4): 3447–3466.

Wilkinson, M.D., M. Dumontier, I.J. Aalbersberg, G. Appleton, M. Axton, A. Baak, N. Blomberg, J.-W. Boiten, et al. 2016. The FAIR guiding principles for scientific data management and stewardship. *Scientific Data* 3(1): Article 160018

Xu, C., J. Wang, Y. Sang, K. Li, J. Liu, and G. Yang. 2023. An effective deep learning model for monitoring mangroves: A case study of the Indus Delta. *Remote Sensing* 15(9): Article 2220

Yamada, T.J., D. Takeuchi, M.A. Farukh, and Y. Kitano. 2016. Climatological characteristics of heavy rainfall in northern Pakistan and atmospheric blocking over western Russia. *Journal of Climate* 29(21): 7743–7754.

Yan, X., and J. Wang. 2022. The forest change footprint of the upper Indus Valley, from 1990 to 2020. *Remote Sensing* 14(3): Article 744.

Yan, X., J. Wang, X. Liu, H. Zhao, and Y. Wu. 2022. Mining the drivers of forest cover change in the upper Indus Valley, high Asia region from 1990 to 2020. *Ecological Indicators* 144: Article 109566